

STATISTICAL PHARMACOLOGY VIA KAKUTANI DICHOTOMY I: THE MARGINAL DRUG KAKUTANI INDEX AND THE $\alpha_c = 1/2$ CRITICALITY THEOREM IN HIGH-DIMENSIONAL CONFORMATIONAL ENSEMBLES

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SUMMARY. This is the first paper of a series that transports the central mechanism of *Localized Bost–Connes systems* [4]—a phase transition that is a tail event of an infinite product, invisible to every finite factor—from arithmetic thermodynamics to the statistical geometry of protein conformational ensembles. We model the conformational space of a protein with N two-state local switches as $\Omega_N = \{0, 1\}^N$ and two ligands L_A, L_B as product measures $P_{A,N} = \bigotimes_i \text{Bernoulli}(p_i)$, $P_{B,N} = \bigotimes_i \text{Bernoulli}(q_i)$. We define the *marginal drug Kakutani index* $K_N(L_A, L_B) = \sum_{i \leq N} h_i$, the sum of the local squared Hellinger distances, the *Kakutani affinity* $\Pi_N = \int_{\Omega_N} \sqrt{dP_{A,N} dP_{B,N}} = \prod_{i \leq N} (1 - h_i/2)$, and the *doubling tail increment* $D_{\text{tail}}(N) = K_{2N} - K_N$. We prove an exact quartic Taylor expansion of h_i in the local shift $\delta_i = q_i - p_i$ (Lemma 2.3), whose odd cubic term vanishes identically at the maximal-entropy point $p = 1/2$, where in fact $h(\delta) = 2 - \sqrt{1 + 2\delta} - \sqrt{1 - 2\delta} = \delta^2 + \frac{5}{4}\delta^4 + \frac{21}{8}\delta^6 + \dots$ exactly. For the power-law allosteric decay model $q_i = p + ci^{-\alpha}$ with $p = 1/2$, $c = 1/10$, we prove the *marginal drug Kakutani phase trichotomy* (Theorem 3.1): with $A = c^2/(4p(1-p)) = 10^{-2}$, one has $K_N \sim AN^{1-2\alpha}/(1-2\alpha)$ for $0 < \alpha < 1/2$, $K_N = A(\ln N + \gamma_{\text{EM}}) + C_\infty(1/2) + O(N^{-1})$ at $\alpha = 1/2$, and $K_N \rightarrow K_\infty(\alpha) < \infty$ for $\alpha > 1/2$; by Kakutani’s theorem the infinite-dimensional ensembles are mutually singular in the first two regimes and equivalent in the third, so $\alpha_c = 1/2$ is the unique critical exponent. At the critical point the doubling increment converges to the universal constant $A \ln 2$ (Proposition 3.2). We then prove (Theorem 4.1) that the effective exponent read from a finite regression window is $\gamma_{\text{eff}}(N; \alpha) = (1 - 2\alpha)/(1 + (1 - 2\alpha)\zeta(2\alpha)N^{2\alpha-1})$, so that the systematic upward bias observed near α_c is the simple pole of ζ at $2\alpha = 1$ and not a numerical error, and that the doubling estimator cancels it identically (Corollary 4.2). At $N = 10^5$ the measured values $K_{10^5} = 6.311486, 0.1211103, 0.0262136, 0.01658727$ for $\alpha = 0.25, 0.5, 0.75, 1$ agree with the Euler–Maclaurin predictions to relative accuracy 10^{-10} or better; the measured $D_{\text{tail}}(10^5)|_{\alpha=1/2} = 0.00693145$ agrees with $A \ln 2 = 0.00693147$ to 3.5×10^{-6} ; and the anomalous regression exponent $\gamma_{\text{num}}(0.45) = 0.1610$ is predicted analytically as 0.1601 at the window centre and 0.1613 on the window itself. The paper closes with an assessment and with the problem left to Paper II: whether the off-diagonal covariance network of a correlated ensemble can drive the Kakutani transition on its own when every marginal is unchanged.

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1. INTRODUCTION: THE BINDING AFFINITY PARADOX AND INFINITE-DIMENSIONAL ENSEMBLE GEOMETRY

The paradox. Consider two ligands L_A and L_B bound to the same target—the presenilin-1 subunit of γ -secretase, or a class A G-protein-coupled receptor—with macroscopically indistinguishable binding free energies,

$$\Delta G_{\text{bind}}(L_A) \approx \Delta G_{\text{bind}}(L_B),$$

and with local conformational footprints that are, residue by residue, almost identical: if p_i and q_i denote the probabilities that the i -th local switch (a side-chain rotamer, a backbone hydrogen bond, a water-mediated contact) is in its “active” state under L_A and under L_B , then $\delta_i = q_i - p_i$ is small for every i , and for the distal residues it tends to zero. The paradox of allosteric drug design [12, 11] is that such a pair may nevertheless drive the protein into *different*—in the extreme case, disjoint—downstream functional states: one ligand is a silent binder, the other a biased agonist or a γ -secretase modulator that shifts the $A\beta_{42}/A\beta_{40}$ ratio.

The classical thermodynamic description [10] cannot resolve the paradox, because it aggregates the ensemble into a handful of global states whose populations are continuous functions of the δ_i ; if every δ_i is small, every population is nearly unchanged. The resolution proposed here is measure-theoretic. The object that governs downstream function is not any finite collection of marginals but the joint conformational law on the high-dimensional cube $\Omega_N = \prod_{i=1}^N \{0, 1\}$, and for that law the question “are P_A and P_B close?” has, in the limit $N \rightarrow \infty$, only two possible answers.

The Kakutani dichotomy. For infinite products of probability measures, Kakutani [8] proved that $\bigotimes_i \mu_i$ and $\bigotimes_i \nu_i$ (with $\mu_i \sim \nu_i$ for each i) are either mutually absolutely continuous or mutually singular, according as $\sum_i \rho(\mu_i, \nu_i)$ converges or diverges, where ρ is the squared Hellinger distance (equivalently, according as the product of the Hellinger affinities $\prod_i \int \sqrt{d\mu_i} d\nu_i$ is positive or zero). There is no third possibility, and the dichotomy is insensitive to any finite number of factors: the decision is made in the tail. The Gaussian analogue is the Feldman–Hájek theorem [5, 6].

This is precisely the structure that organises [4], the localized generalisation of the Bost–Connes system [3]. There the KMS_β simplex of the S -local Bost–Connes system is $\text{Prob}(G_S/\Xi_\beta^\perp)$,

where the obstruction group

$$\Xi_\beta(K, S) = \left\{ \chi \in \widehat{G_S} : \sum_{\mathfrak{p} \in S} N\mathfrak{p}^{-\beta} |1 - \chi(\sigma_{\mathfrak{p}})|^2 < \infty \right\}$$

is extracted from a Kakutani product criterion [4, Ch. 1, Props. 3.3–3.4 and Thm. 3.6], and the central negative lesson of that book is that divergence of the partition function $\zeta_{K,S}(\beta) = \infty$ does *not* imply uniqueness of the KMS_β state: the transition is “a tail event—a Kakutani dichotomy in which every single factor is harmless, and only the infinite product degenerates” [4, Preface]. The dictionary between that setting and the present one is given below; it is exact at the level of the criterion, and it is the reason we call the series *statistical pharmacology via Kakutani dichotomy*.

THE DICTIONARY between the localized Bost–Connes classification [4] and the marginal drug Kakutani index.

	Localized Bost–Connes system	Conformational ensemble
Index set	primes $\mathfrak{p} \in S$	local switches $i = 1, \dots, N$
Local weight	$N\mathfrak{p}^{-\beta} 1 - \chi(\sigma_{\mathfrak{p}}) ^2$	h_i , the squared Hellinger distance of Bernoulli(p_i), Bernoulli(q_i)
Tail functional	$\sum_{\mathfrak{p} \in S} N\mathfrak{p}^{-\beta} 1 - \chi(\sigma_{\mathfrak{p}}) ^2$	$K_N = \sum_{i \leq N} h_i$
Control parameter	inverse temperature β	allosteric decay exponent α
Convergent regime	$\chi \in \Xi_\beta$, symmetry broken	$P_A \sim P_B$: localized perturbation
Divergent regime	$\chi \notin \Xi_\beta$, state unique	$P_A \perp P_B$: global allosteric re-shaping
False expectation	$\zeta_{K,S}(\beta) = \infty \Rightarrow$ uniqueness	$\delta_i \rightarrow 0 \Rightarrow P_A \approx P_B$

Main results. Section 2 defines the product conformational ensemble, the marginal drug Kakutani index K_N , the Kakutani affinity Π_N and the doubling tail increment $D_{\text{tail}}(N)$, and proves the exact quartic expansion of the local Hellinger coordinate (Lemma 2.3); at $p = 1/2$ the expansion is even and is given in closed form. Section 3 proves the phase trichotomy at $\alpha_c = 1/2$ for the power-law model $q_i = p + c i^{-\alpha}$ (Theorem 3.1), including the constants, and the universal doubling identity $D_{\text{tail}} \rightarrow A \ln 2$ at criticality (Proposition 3.2). Section 4 develops the Euler–Maclaurin finite-size scaling theory: Theorem 4.1 gives the closed-form effective exponent $\gamma_{\text{eff}}(N; \alpha)$ and identifies the near-critical bias with the pole of $\zeta(2\alpha)$; Corollary 4.2 shows that the doubling estimator is unbiased. Section 5 tabulates the numerical experiment of `exp01_marginal_dki_criticality.py` (Tables 1 and 2, Figure 1) against the analytic predictions. Section 6 separates what has been established from what has not, and formulates the covariance problem for Paper II.

Standing notation. Throughout, $p \in (0, 1)$ is the ground-state activation probability, $c \in (0, \min(p, 1 - p))$ the perturbation amplitude, $\alpha > 0$ the decay exponent, and

$$A = A(p, c) = \frac{c^2}{4p(1-p)}, \quad H_N(s) = \sum_{i=1}^N i^{-s} \quad (s \in \mathbb{R}), \quad \gamma_{\text{EM}} = 0.5772156649 \dots \quad (1.1)$$

denote the prefactor, the generalised harmonic numbers and the Euler–Mascheroni constant. ζ is the Riemann zeta function, meromorphic on $\mathbb{C}$ with a simple pole at 1 of residue 1. In the numerical sections $p = 1/2$, $c = 1/10$, so $A = 10^{-2}$.

2. PRODUCT CONFORMATIONAL SPACES AND THE MARGINAL DRUG KAKUTANI INDEX

Definition 2.1 (Product conformational ensemble). Let a protein carry N local two-state conformational switches, and let $\Omega_N = \{0, 1\}^N$ with coordinates $X = (X_1, \dots, X_N)$. Given activation profiles $(p_i)_{i \leq N}$ and $(q_i)_{i \leq N}$ in $(0, 1)$ induced by ligands L_A and L_B , the *marginal (first-order) conformational ensembles* are the product measures

$$P_{A,N} = \bigotimes_{i=1}^N \text{Bernoulli}(p_i), \quad P_{B,N} = \bigotimes_{i=1}^N \text{Bernoulli}(q_i), \quad (2.1)$$

where $\text{Bernoulli}(r)$ is the law on $\{0, 1\}$ with $\text{Bernoulli}(r)\{1\} = r$. For infinite profiles $(p_i)_{i \geq 1}$, $(q_i)_{i \geq 1}$ the same formulas with $N = \infty$ define P_A , P_B on $\Omega_\infty = \{0, 1\}^\mathbb{N}$ with the product σ -algebra.

Definition 2.2 (Local Hellinger coordinate, cumulative DKI, Kakutani affinity, doubling tail). For $i \geq 1$ let

$$h_i = h(p_i, q_i) = (\sqrt{p_i} - \sqrt{q_i})^2 + (\sqrt{1 - p_i} - \sqrt{1 - q_i})^2 \quad (2.2)$$

be the squared Hellinger distance between $\text{Bernoulli}(p_i)$ and $\text{Bernoulli}(q_i)$. The *marginal drug Kakutani index* (DKI), the *Kakutani affinity*, and the *doubling tail increment* are

$$K_N(L_A, L_B) = \sum_{i=1}^N h_i, \quad \Pi_N(L_A, L_B) = \int_{\Omega_N} \sqrt{dP_{A,N} dP_{B,N}}, \quad D_{\text{tail}}(N) = K_{2N} - K_N. \quad (2.3)$$

Lemma 2.3 (Exact quartic Taylor expansion of the local Hellinger distance). *Let $p \in (0, 1)$ and $q = p + \delta$ with $|\delta| < \min(p, 1 - p)$. Then*

$$h(p, p + \delta) = \frac{\delta^2}{4p(1-p)} - \frac{(1-2p)\delta^3}{8p^2(1-p)^2} + \frac{5(1-3p+3p^2)\delta^4}{64p^3(1-p)^3} + O(\delta^5), \quad (2.4)$$

the series being convergent for $|\delta| < \min(p, 1 - p)$. In particular, at the maximal-entropy symmetric point $p = 1/2$ the odd terms vanish identically, and in fact

$$h(\tfrac{1}{2}, \tfrac{1}{2} + \delta) = 2 - \sqrt{1+2\delta} - \sqrt{1-2\delta} = \sum_{k \geq 1} a_k \delta^{2k}, \quad a_k = -2 \binom{1/2}{2k} 4^k > 0, \quad (2.5)$$

with $a_1 = 1$, $a_2 = \frac{5}{4}$, $a_3 = \frac{21}{8}$, $a_4 = \frac{429}{64}$, and the series converges for $|\delta| < 1/2$. Thus

$$h(\tfrac{1}{2}, \tfrac{1}{2} + \delta) = \delta^2 + \frac{5}{4}\delta^4 + \frac{21}{8}\delta^6 + O(\delta^8). \quad (2.6)$$

Proof. Write $\sqrt{p+\delta} = \sqrt{p}(1+u)^{1/2}$ with $u = \delta/p$, $|u| < 1$, and expand by the binomial series,

$$(1+u)^{1/2} = 1 + \frac{u}{2} - \frac{u^2}{8} + \frac{u^3}{16} - \frac{5u^4}{128} + \frac{7u^5}{256} - \dots, \quad \binom{1/2}{n} = \frac{(-1)^{n-1}(2n-3)!!}{2^n n!} \quad (n \geq 1).$$

Hence $\sqrt{p} - \sqrt{p+\delta} = -\sqrt{p}(\frac{u}{2} - \frac{u^2}{8} + \frac{u^3}{16} - \frac{5u^4}{128} + O(u^5))$ and, squaring,

$$\begin{aligned} (\sqrt{p} - \sqrt{p+\delta})^2 &= p \left[\frac{u^2}{4} - 2 \cdot \frac{u}{2} \cdot \frac{u^2}{8} + \left(\frac{u^4}{64} + 2 \cdot \frac{u}{2} \cdot \frac{u^3}{16} \right) + O(u^5) \right] \\ &= p \left[\frac{u^2}{4} - \frac{u^3}{8} + \frac{5u^4}{64} + O(u^5) \right] = \frac{\delta^2}{4p} - \frac{\delta^3}{8p^2} + \frac{5\delta^4}{64p^3} + O(\delta^5). \end{aligned}$$

The second summand of (2.2) is obtained from the first by the substitution $(p, \delta) \mapsto (1-p, -\delta)$:

$$(\sqrt{1-p} - \sqrt{1-p-\delta})^2 = \frac{\delta^2}{4(1-p)} + \frac{\delta^3}{8(1-p)^2} + \frac{5\delta^4}{64(1-p)^3} + O(\delta^5).$$

Adding, the quadratic coefficient is $\frac{1}{4}(\frac{1}{p} + \frac{1}{1-p}) = \frac{1}{4p(1-p)}$; the cubic coefficient is

$$\frac{1}{8} \left(\frac{1}{(1-p)^2} - \frac{1}{p^2} \right) = \frac{p^2 - (1-p)^2}{8p^2(1-p)^2} = \frac{2p-1}{8p^2(1-p)^2} = -\frac{1-2p}{8p^2(1-p)^2};$$

and the quartic coefficient is

$$\frac{5}{64} \left(\frac{1}{p^3} + \frac{1}{(1-p)^3} \right) = \frac{5((1-p)^3 + p^3)}{64p^3(1-p)^3} = \frac{5(1-3p+3p^2)}{64p^3(1-p)^3},$$

since $(1-p)^3 + p^3 = 1 - 3p + 3p^2$. This is (2.4); convergence follows from that of the two binomial series. At $p = 1/2$ the cubic coefficient carries the factor $1 - 2p = 0$. More generally,

at $p = 1/2$ the two summands of (2.2) are exchanged by $\delta \mapsto -\delta$, so h is an even function of δ and *all* odd coefficients vanish. For the closed form, expand the squares in (2.2):

$$h(p, q) = (p+q) - 2\sqrt{pq} + (2-p-q) - 2\sqrt{(1-p)(1-q)} = 2 - 2\left(\sqrt{pq} + \sqrt{(1-p)(1-q)}\right), \quad (2.7)$$

and at $p = 1/2$, $q = 1/2 + \delta$,

$$2\sqrt{pq} = 2\sqrt{\frac{1}{2}\left(\frac{1}{2} + \delta\right)} = \sqrt{1 + 2\delta}, \quad 2\sqrt{(1-p)(1-q)} = \sqrt{1 - 2\delta},$$

whence $h = 2 - \sqrt{1 + 2\delta} - \sqrt{1 - 2\delta}$. With $x = 2\delta$ and $\sqrt{1 \pm x} = \sum_n \binom{1/2}{n} (\pm x)^n$, the odd powers cancel and $h = -2 \sum_{k \geq 1} \binom{1/2}{2k} x^{2k} = \sum_{k \geq 1} (-2 \binom{1/2}{2k} 4^k) \delta^{2k}$. Since $\binom{1/2}{2k} < 0$ for $k \geq 1$, all $a_k > 0$; the values $a_1 = -2 \cdot (-\frac{1}{8}) \cdot 4 = 1$, $a_2 = -2 \cdot (-\frac{5}{128}) \cdot 16 = \frac{5}{4}$, $a_3 = -2 \cdot (-\frac{21}{1024}) \cdot 64 = \frac{21}{8}$, $a_4 = -2 \cdot (-\frac{429}{32768}) \cdot 256 = \frac{429}{64}$ follow from $\binom{1/2}{2} = -\frac{1}{8}$, $\binom{1/2}{4} = -\frac{5}{128}$, $\binom{1/2}{6} = -\frac{21}{1024}$, $\binom{1/2}{8} = -\frac{429}{32768}$. Consistency with (2.4) at $p = 1/2$: $\frac{5(1-\frac{3}{2}+\frac{3}{4})}{64 \cdot \frac{1}{64}} = \frac{5}{4}$. $\square$

Remark 2.4 (Fisher information and the quality of the quadratic term). The quadratic coefficient $1/(4p(1-p))$ is one quarter of the Fisher information $I(p) = 1/(p(1-p))$ of the Bernoulli family, in accordance with the general identity $h(P_\theta, P_{\theta+\delta}) = \frac{1}{4}I(\theta)\delta^2 + O(\delta^3)$ [9]. At $p = 1/2$ the absence of the cubic term makes the quadratic approximation accurate to fourth order: the relative quartic correction at the largest shift $\delta_1 = c = 1/10$ is $a_2 c^2 = \frac{5}{4} \times 10^{-2} = 1.25\%$, and the exact value is $h(\frac{1}{2}, \frac{6}{10}) = 2 - \sqrt{1.2} - \sqrt{0.8} = 0.0101276940\dots$, against $\delta^2 = 0.01$ and $\delta^2 + \frac{5}{4}\delta^4 = 0.010125$. For $i \geq 2$ the shift is smaller and the correction is smaller still ($a_2 c^2 i^{-2\alpha}$). Since all $a_k > 0$, the function $\delta \mapsto h(\frac{1}{2}, \frac{1}{2} + \delta)/\delta^2$ is increasing in $|\delta|$, so for the model of Section 3 we have the two-sided bound

$$\delta_i^2 \leq h_i \leq (1 + \varepsilon_c) \delta_i^2, \quad \varepsilon_c := \frac{h(\frac{1}{2}, \frac{1}{2} + c)}{c^2} - 1 = 0.0127694\dots \quad (c = \frac{1}{10}), \quad (2.8)$$

which is what the qualitative part of the trichotomy needs.

Lemma 2.5 (Product formula for the affinity). *For every N ,*

$$\Pi_N = \prod_{i=1}^N \left(\sqrt{p_i q_i} + \sqrt{(1-p_i)(1-q_i)} \right) = \prod_{i=1}^N \left(1 - \frac{h_i}{2} \right), \quad (2.9)$$

and $0 < 1 - h_i/2 \leq 1$ with equality iff $p_i = q_i$. Moreover

$$\ln \Pi_N = -\frac{K_N}{2} - Q_N, \quad 0 \leq Q_N \leq \frac{1}{4} \sum_{i=1}^N h_i^2. \quad (2.10)$$

Proof. On the finite set Ω_N the integral $\int \sqrt{dP_{A,N} dP_{B,N}}$ is the sum over $x \in \Omega_N$ of $\sqrt{P_{A,N}\{x\} P_{B,N}\{x\}}$. Since $P_{A,N}\{x\} = \prod_i p_i^{x_i} (1-p_i)^{1-x_i}$ and likewise for $P_{B,N}$, the square root factorises and the sum over x factorises into $\prod_i (\sqrt{p_i q_i} + \sqrt{(1-p_i)(1-q_i)})$. By the identity (2.7) from the proof of Lemma 2.3, $\sqrt{p_i q_i} + \sqrt{(1-p_i)(1-q_i)} = 1 - h_i/2$, which is the second equality in (2.9). By Cauchy-Schwarz, $\sqrt{p_i q_i} + \sqrt{(1-p_i)(1-q_i)} \leq \sqrt{(p_i + 1-p_i)(q_i + 1-q_i)} = 1$ with equality iff $(\sqrt{p_i}, \sqrt{1-p_i})$ and $(\sqrt{q_i}, \sqrt{1-q_i})$ are proportional, i.e. iff $p_i = q_i$; positivity is clear. For (2.10) put $x_i = h_i/2 \in (0, 1/2]$ (indeed $h_i < 2$ because $\sqrt{p_i q_i} > 0$). The series $-\ln(1-x) = \sum_{m \geq 1} x^m/m$ gives $-\ln(1-x) - x = \sum_{m \geq 2} x^m/m \in [0, x^2 \sum_{m \geq 0} 2^{-m}/2] = [0, x^2]$ for $0 \leq x \leq 1/2$. Summing over i with $Q_N := \sum_i (-\ln(1-x_i) - x_i)$ yields (2.10). $\square$

Theorem 2.6 (Kakutani). *Let $(p_i), (q_i) \subset (0, 1)$ and P_A, P_B be as in Definition 2.1. Then exactly one of the following holds:*

- (i) $\sum_{i \geq 1} h_i < \infty$; then $\Pi_\infty := \lim_N \Pi_N > 0$ and $P_A \sim P_B$ (mutually absolutely continuous);
- (ii) $\sum_{i \geq 1} h_i = \infty$; then $\Pi_\infty = 0$ and $P_A \perp P_B$ (mutually singular).

Proof. This is Kakutani's theorem [8] for the factors $\mu_i = \text{Bernoulli}(p_i)$, $\nu_i = \text{Bernoulli}(q_i)$, which are equivalent since $p_i, q_i \in (0, 1)$; the criterion there is $\prod_i \int \sqrt{d\mu_i d\nu_i} > 0$ versus $= 0$, and by Lemma 2.5 $\prod_i (1 - h_i/2) > 0$ iff $\sum_i h_i < \infty$ (because $0 < h_i/2 \leq 1/2$ and $\ln(1-x) \asymp -x$ on $[0, 1/2]$). $\square$

3. THE MARGINAL DRUG KAKUTANI PHASE TRICHOTOMY AT $\alpha_c = 1/2$

Standing assumption: power-law allosteric decay. Throughout this section and the next, $p_i = p = \frac{1}{2}$ for all i , and $q_i = p + ci^{-\alpha}$ with $c = \frac{1}{10}$ and $\alpha > 0$. Thus $\delta_i = ci^{-\alpha} \in (0, c]$, all $q_i \in (\frac{1}{2}, \frac{3}{5}]$, and $A = c^2/(4p(1-p)) = c^2 = 10^{-2}$. With a_k the coefficients of (2.5) we write

$$R_N(\alpha) := \sum_{k \geq 2} a_k c^{2k} H_N(2k\alpha), \quad C_\infty(\alpha) := \sum_{k \geq 2} a_k c^{2k} \zeta(2k\alpha) \quad (\alpha > \frac{1}{4}) \quad (3.1)$$

for the higher-order Hellinger corrections; Lemma 3.4 shows that $C_\infty(\alpha) = \lim_N R_N(\alpha)$. We state the two main results of the section first; their proofs follow the two lemmas.

Theorem 3.1 (Marginal drug Kakutani phase trichotomy). *Under the standing assumption, with $A = 10^{-2}$ and $R_N(\alpha)$, $C_\infty(\alpha)$ as in (3.1), the following hold as $N \rightarrow \infty$.*

(1) **Supercritical allosteric phase**, $0 < \alpha < \frac{1}{2}$.

$$K_N(\alpha) = \frac{A}{1-2\alpha} N^{1-2\alpha} + A\zeta(2\alpha) + \frac{A}{2} N^{-2\alpha} + R_N(\alpha) + O(N^{-2\alpha-1}) \rightarrow \infty, \quad (3.2)$$

where $R_N(\alpha) = C_\infty(\alpha) + O(N^{1-4\alpha})$ if $\alpha > \frac{1}{4}$ and $R_N(\alpha) = O(N^{1-4\alpha})$ (resp. $O(\ln N)$) if $\alpha < \frac{1}{4}$ (resp. $= \frac{1}{4}$). Consequently $\ln \Pi_N = -\frac{1}{2} K_N(\alpha) + O(1)$ for $\alpha > \frac{1}{4}$ and $\ln \Pi_N = -\frac{1}{2} K_N(\alpha)(1 + o(1))$ in general; $\Pi_N \rightarrow 0$ and $P_A \perp P_B$.

(2) **Critical Kakutani boundary**, $\alpha = \alpha_c = \frac{1}{2}$.

$$\begin{aligned} K_N(\tfrac{1}{2}) &= A \left(\ln N + \gamma_{\text{EM}} + \frac{1}{2N} \right) + C_\infty(\tfrac{1}{2}) + O(c^4 N^{-1}) \rightarrow \infty, \\ C_\infty(\tfrac{1}{2}) &= \sum_{k \geq 2} a_k c^{2k} \zeta(k) = 2.0885 \times 10^{-4}, \end{aligned} \quad (3.3)$$

and

$$\Pi_N = \kappa N^{-A/2} (1 + O(N^{-1})), \quad \kappa = \exp\left(-\frac{1}{2}(A\gamma_{\text{EM}} + C_\infty(\tfrac{1}{2})) - Q_\infty\right) \in (0, 1), \quad (3.4)$$

so $\Pi_N \rightarrow 0$ by a critical power-law collapse and $P_A \perp P_B$.

(3) **Subcritical localized phase**, $\alpha > \frac{1}{2}$.

$$\begin{aligned} K_\infty(\alpha) &:= \lim_{N \rightarrow \infty} K_N(\alpha) = A\zeta(2\alpha) + C_\infty(\alpha) < \infty, \\ K_N(\alpha) &= K_\infty(\alpha) - A\zeta(2\alpha, N+1) - O(c^4 N^{1-4\alpha}), \end{aligned} \quad (3.5)$$

with $A\zeta(2\alpha, N+1) = \frac{A}{2\alpha-1} N^{1-2\alpha} (1 + O(N^{-1}))$; $\Pi_\infty(\alpha) = \lim_N \Pi_N > 0$ and $P_A \sim P_B$.

In particular $\alpha_c = \frac{1}{2}$ is the unique critical exponent: the infinite-dimensional ensembles are mutually singular for $\alpha \leq \frac{1}{2}$ and equivalent for $\alpha > \frac{1}{2}$.

Proposition 3.2 (Scale-invariant doubling tail identity at $\alpha_c = 1/2$). *Under the standing assumption at $\alpha = \frac{1}{2}$,*

$$\begin{aligned} D_{\text{tail}}(N) &= K_{2N} - K_N = A \ln 2 - \frac{A}{4N} + O(c^4 N^{-1}) + O(N^{-2}), \\ \lim_{N \rightarrow \infty} D_{\text{tail}}(N) &= A \ln 2 = 0.0069314718 \dots \end{aligned} \quad (3.6)$$

The limit is independent of N and of the Euler–Mascheroni and C_∞ constants, which cancel identically.

We first record the Euler–Maclaurin expansion of the generalised harmonic numbers, with the constant identified.

Lemma 3.3 (Euler–Maclaurin expansion of $H_N(s)$). *For $s > 0$, $s \neq 1$, as $N \rightarrow \infty$,*

$$H_N(s) = \frac{N^{1-s}}{1-s} + \zeta(s) + \frac{N^{-s}}{2} + \frac{s N^{-s-1}}{12} + O(N^{-s-3}), \quad (3.7)$$

and for $s = 1$,

$$H_N(1) = \ln N + \gamma_{\text{EM}} + \frac{1}{2N} - \frac{1}{12N^2} + O(N^{-4}). \quad (3.8)$$

For $s > 1$ one has the exact tail identity $\zeta(s) - H_N(s) = \sum_{i>N} i^{-s} = \zeta(s, N+1)$ (Hurwitz zeta), and $\frac{(N+1)^{1-s}}{s-1} \leq \zeta(s, N+1) \leq \frac{N^{1-s}}{s-1}$.

Proof. Apply the Euler–Maclaurin summation formula [1] to $f(x) = x^{-s}$ on $[1, N]$ with two Bernoulli terms:

$$\begin{aligned} \sum_{i=1}^N f(i) &= \int_1^N f(x) dx + \frac{f(1) + f(N)}{2} + \frac{B_2}{2!} (f'(N) - f'(1)) + R_2(N), \\ R_2(N) &= - \int_1^N \frac{\overline{B}_3(x)}{3!} f'''(x) dx, \end{aligned}$$

where $B_2 = \frac{1}{6}$, $\overline{B}_3$ is the periodic Bernoulli function, and the remainder can be pushed to any order. Since $f'''(x) = -s(s+1)(s+2)x^{-s-3}$ is absolutely integrable at ∞ for $s > 0$, $R_2(N) = R_2(\infty) + O(N^{-s-2})$, and pushing one more step, $R_2(N) = R_2(\infty) + \frac{B_4}{4!} f''''(N) + O(N^{-s-4})$ with $B_4 = -\frac{1}{30}$, which is $O(N^{-s-3})$. For $s \neq 1$, $\int_1^N x^{-s} dx = \frac{N^{1-s}-1}{1-s}$ and $f'(N) = -sN^{-s-1}$, so

$$\begin{aligned} H_N(s) &= \frac{N^{1-s}}{1-s} + C(s) + \frac{N^{-s}}{2} + \frac{sN^{-s-1}}{12} + O(N^{-s-3}), \\ C(s) &:= -\frac{1}{1-s} + \frac{1}{2} + \frac{s}{12} + R_2(\infty). \end{aligned}$$

For $s > 1$ the term $N^{1-s}/(1-s) \rightarrow 0$, so $C(s) = \lim_N H_N(s) = \zeta(s)$. The function $C(s)$ defined by the displayed formula is analytic in $\text{Re } s > -2$, $s \neq 1$ (the remainder integral converges there), and $C(s) = \zeta(s)$ for $s > 1$; by analytic continuation $C(s) = \zeta(s)$ for all $s > 0$, $s \neq 1$ [13, §2.1, eq. (2.1.4)]. This is (3.7). For $s = 1$ the same computation with $\int_1^N x^{-1} dx = \ln N$ gives $H_N(1) = \ln N + C(1) + \frac{1}{2N} - \frac{1}{12N^2} + O(N^{-4})$, and $C(1) = \lim_N (H_N(1) - \ln N) = \gamma_{\text{EM}}$ by definition of the Euler–Mascheroni constant. The tail identity for $s > 1$ is the definition of the Hurwitz zeta function, and the bracketing bounds follow from comparing $\sum_{i>N} i^{-s}$ with $\int_N^\infty x^{-s} dx$ and $\int_{N+1}^\infty x^{-s} dx$, the summand being decreasing. $\square$

Lemma 3.4 (Series representation of the DKI). *Under the standing assumption, for every $N \geq 1$,*

$$K_N(\alpha) = \sum_{k \geq 1} a_k c^{2k} H_N(2k\alpha) = A H_N(2\alpha) + R_N(\alpha), \quad (3.9)$$

with a_k as in (2.5) and $R_N(\alpha)$ as in (3.1), the double series converging absolutely. Moreover $0 < R_N(\alpha) \leq \varepsilon_c A H_N(4\alpha)$, so that

$$R_N(\alpha) = \begin{cases} O(1), & \alpha > \frac{1}{4}, \\ O(\ln N), & \alpha = \frac{1}{4}, \\ O(N^{1-4\alpha}), & \alpha < \frac{1}{4}. \end{cases} \quad (3.10)$$

In every case $R_N(\alpha) = o(N^{1-2\alpha})$ when $\alpha < \frac{1}{2}$, and for $\alpha > \frac{1}{4}$

$$\lim_{N \rightarrow \infty} R_N(\alpha) = \sum_{k \geq 2} a_k c^{2k} \zeta(2k\alpha) = C_\infty(\alpha), \quad (3.11)$$

with $R_N(\alpha) = C_\infty(\alpha) - a_2 c^4 \zeta(4\alpha, N+1) - \dots = C_\infty(\alpha) + O(N^{1-4\alpha})$.

Proof. By Lemma 2.3, $h_i = \sum_k a_k \delta_i^{2k} = \sum_k a_k c^{2k} i^{-2k\alpha}$, with all terms positive and $\delta_i \leq c < \frac{1}{2}$; summing over $i \leq N$ and interchanging the finite sum with the convergent positive series gives (3.9). For the bound, $R_N = \sum_i (h_i - \delta_i^2)$ and, since all $a_k > 0$ and $\delta_i \leq c$, $h_i - \delta_i^2 = \delta_i^4 \sum_{k \geq 2} a_k \delta_i^{2k-4} \leq \delta_i^4 \sum_{k \geq 2} a_k c^{2k-4} = \delta_i^4 (h(\frac{1}{2}, \frac{1}{2} + c) - c^2)/c^4 = \varepsilon_c \delta_i^4/c^2$, whence $R_N \leq \varepsilon_c c^{-2} \sum_i c^4 i^{-4\alpha} = \varepsilon_c A H_N(4\alpha)$. The growth statements (3.10) are Lemma 3.3 applied to $H_N(4\alpha)$, and $N^{1-4\alpha} = o(N^{1-2\alpha})$, $\ln N = o(N^{1/2})$. For $\alpha > 1/4$ each $H_N(2k\alpha)$ with $k \geq 2$ increases to $\zeta(2k\alpha) < \infty$ and dominated convergence (dominating series $\sum_k a_k c^{2k} \zeta(2k\alpha) \leq \sum_k a_k c^{2k} \zeta(4\alpha) < \infty$) gives (3.11); the rate follows from the tail identity of Lemma 3.3 applied termwise, the $k = 2$ term dominating. $\square$

Proof of Theorem 3.1. Qualitative part. By (2.8), $A H_N(2\alpha) \leq K_N(\alpha) \leq (1 + \varepsilon_c) A H_N(2\alpha)$, and $H_N(2\alpha)$ converges iff $2\alpha > 1$. Hence $\sum_i h_i < \infty$ iff $\alpha > \frac{1}{2}$, and Theorem 2.6 gives $P_A \sim P_B$ for $\alpha > \frac{1}{2}$ and $P_A \perp P_B$ for $\alpha \leq \frac{1}{2}$, together with $\Pi_\infty > 0$ resp. $\Pi_\infty = 0$.

(1). Insert (3.7) with $s = 2\alpha \in (0, 1)$ into $K_N = A H_N(2\alpha) + R_N$ (Lemma 3.4); the term $\frac{A s}{12} N^{-s-1}$ is absorbed into $O(N^{-2\alpha-1})$. The description of R_N is (3.10)–(3.11). For Π_N use (2.10): $Q_N \leq \frac{1}{4} \sum_i h_i^2 \leq \frac{1}{4} (1 + \varepsilon_c)^2 c^4 H_N(4\alpha)$, which is bounded for $\alpha > \frac{1}{4}$ and $O(N^{1-4\alpha}) = o(N^{1-2\alpha})$ otherwise.

(2). At $\alpha = \frac{1}{2}$ the k -th term of (3.9) is $a_k c^{2k} H_N(k)$. For $k = 1$ use (3.8): $A H_N(1) = A(\ln N + \gamma_{\text{EM}} + \frac{1}{2N}) + O(N^{-2})$. For $k \geq 2$, $H_N(k) = \zeta(k) - \zeta(k, N+1)$ with $0 < \zeta(k, N+1) \leq N^{1-k}/(k-1) \leq N^{-1}$, so $\sum_{k \geq 2} a_k c^{2k} H_N(k) = C_\infty(\frac{1}{2}) - \theta_N$, $0 < \theta_N \leq N^{-1} \sum_{k \geq 2} a_k c^{2k} = \varepsilon_c A N^{-1}$; this is the $O(c^4 N^{-1})$ term. The numerical value of $C_\infty(\frac{1}{2})$ is computed from $a_2 c^4 \zeta(2) = 1.25 \times 10^{-4} \times 1.6449341 = 2.05617 \times 10^{-4}$, $a_3 c^6 \zeta(3) = 2.625 \times 10^{-6} \times 1.2020569 = 3.1554 \times 10^{-6}$, $a_4 c^8 \zeta(4) = 6.7031 \times 10^{-8} \times 1.0823232 = 7.255 \times 10^{-8}$, and $a_5 c^{10} \zeta(5) = 1.97 \times 10^{-9}$, the remaining terms being below 10^{-10} : $C_\infty(\frac{1}{2}) = 2.08847 \times 10^{-4}$. For (3.4), by (2.10), $\ln \Pi_N = -\frac{1}{2} K_N - Q_N$ with Q_N increasing to $Q_\infty \leq \frac{1}{4} (1 + \varepsilon_c)^2 c^4 \zeta(2) = 4.22 \times 10^{-5}$ and $Q_\infty - Q_N \leq \frac{1}{4} (1 + \varepsilon_c)^2 c^4 \zeta(2, N+1) = O(N^{-1})$; inserting (3.3) gives $\ln \Pi_N = -\frac{A}{2} \ln N - \frac{1}{2} (A \gamma_{\text{EM}} + C_\infty(\frac{1}{2})) - Q_\infty + O(N^{-1})$.

(3). For $\alpha > \frac{1}{2}$ every $H_N(2k\alpha)$ increases to $\zeta(2k\alpha)$, and the limit (3.5) follows from monotone convergence in (3.9). The finite- N correction is

$$K_\infty - K_N = A \zeta(2\alpha, N+1) + \sum_{k \geq 2} a_k c^{2k} \zeta(2k\alpha, N+1);$$

by the tail bounds of Lemma 3.3 the k -th term is at most $a_k c^{2k} N^{1-2k\alpha}/(2k\alpha - 1)$, and the sum over $k \geq 2$ is $O(c^4 N^{1-4\alpha})$. The stated form of $A \zeta(2\alpha, N+1)$ is (3.7) for $s = 2\alpha > 1$ combined with the tail identity. $\square$

Remark 3.5 (Numerical confirmation of the constants at $N = 10^5$). The analytic expressions of Theorem 3.1 reproduce the measured values of Table 1 digit by digit.

- *Critical point.* $A(\ln 10^5 + \gamma_{\text{EM}} + \frac{1}{2 \cdot 10^5}) = 10^{-2} \times (11.51292546 + 0.57721566 + 0.00000500) = 0.1209014613$. Adding $C_\infty(\frac{1}{2}) = 0.0002088470$ gives 0.1211103083; the measured value is $K_{10^5}(\frac{1}{2}) = 0.1211103068$, the difference 1.5×10^{-9} being the $-\theta_N = O(c^4 N^{-1})$ truncation of the $k \geq 2$ terms. The leading logarithm accounts for 99.83% of the value; the quartic Hellinger correction $a_2 c^4 \zeta(2)$ for 0.17%; everything beyond that for 3×10^{-6} relative.
- $\alpha = 0.75$. $A \zeta(1.5) = 10^{-2} \times 2.6123753 = 0.0261237535$ and $C_\infty(0.75) = a_2 c^4 \zeta(3) + a_3 c^6 \zeta(4.5) + \dots = 1.50257 \times 10^{-4} + 2.769 \times 10^{-6} + 6.8 \times 10^{-8} = 1.53095 \times 10^{-4}$, so $K_\infty(0.75) = 0.0262768494$. The truncation $A \zeta(1.5, 10^5 + 1) = 6.32454 \times 10^{-5}$ (its leading form $2AN^{-1/2} = 6.32456 \times 10^{-5}$) gives $K_{10^5}(0.75) = 0.0262136040$; measured: 0.0262136040.
- $\alpha = 1$. $A \zeta(2) = 10^{-2} \times \pi^2/6 = 0.0164493407$, $C_\infty(1) = a_2 c^4 \zeta(4) + a_3 c^6 \zeta(6) + a_4 c^8 \zeta(8) + \dots = 1.35290 \times 10^{-4} + 2.6705 \times 10^{-6} + 6.73 \times 10^{-8} = 1.38030 \times 10^{-4}$, so $K_\infty(1) = 0.0165873709$; the truncation $A \zeta(2, 10^5 + 1) = 9.99995 \times 10^{-8} \approx A/N$ gives $K_{10^5}(1) = 0.0165872709$; measured: 0.0165872709.

- $\alpha = 0.25$. $A(2\sqrt{N} + \zeta(\frac{1}{2}) + \frac{1}{2\sqrt{N}}) = 10^{-2} \times (632.4555320 - 1.4603545 + 0.0015811) = 6.3099676$; here $R_N(\frac{1}{4}) = a_2 c^4 H_N(1) + a_3 c^6 H_N(\frac{3}{2}) + \dots = 1.25 \times 10^{-4} \times 12.0901461 + 2.625 \times 10^{-6} \times 2.6060 + \dots = 0.0015182$ is logarithmically growing, in accordance with (3.10); the total 6.3114858 matches the measured 6.3114858.

Proof of Proposition 3.2. By (3.3) applied at $2N$ and at N ,

$$\begin{aligned} K_{2N} - K_N &= A \left(\ln 2N - \ln N + \frac{1}{4N} - \frac{1}{2N} \right) + (C_\infty(\tfrac{1}{2}) - C_\infty(\tfrac{1}{2})) + O(c^4 N^{-1}) + O(N^{-2}) \\ &= A \ln 2 - \frac{A}{4N} + O(c^4 N^{-1}) + O(N^{-2}), \end{aligned}$$

where the $O(N^{-2})$ collects the $-\frac{1}{12N^2}$ terms of (3.8). The $O(c^4 N^{-1})$ term is $\theta_N - \theta_{2N}$, which is at most $\varepsilon_c A N^{-1} = 1.3 \times 10^{-4} A/N$ in absolute value and is therefore numerically dominated by $-A/(4N)$. $\square$

Remark 3.6. Table 1 gives $D_{\text{tail}}(10^5)|_{\alpha=1/2} = 0.0069314474$; the prediction $A \ln 2 - A/(4N) = 0.0069314718 - 0.0000000250 = 0.0069314468$ is off by 6×10^{-10} , i.e. the θ -terms, and the relative deviation from the universal limit $A \ln 2$ is 3.5×10^{-6} . Reading down the $\alpha = 0.5$ block of Table 1, $D_{\text{tail}}(N)$ increases monotonically to $A \ln 2$ from below, exactly as the sign of $-A/(4N)$ dictates.

4. EULER–MACLAURIN FINITE-SIZE SCALING THEORY AND RESOLUTION OF THE NEAR-CRITICAL ANOMALY

The numerical experiment estimates the exponent of K_N by ordinary least squares of $\ln K_N$ on $\ln N$ over a log-uniform grid in the window $[10^3, 10^5]$ (sixty points), and reports $\gamma_{\text{num}}(\alpha)$ against the asymptotic value $\gamma_{\text{th}} = 1 - 2\alpha$. Table 2 shows that the agreement is excellent for $\alpha \leq 0.25$ and degrades steadily as $\alpha \uparrow \frac{1}{2}$, the discrepancy reaching 0.061 at $\alpha = 0.45$. We now prove that this is the signature of the pole of $\zeta(s)$ at $s = 1$ and not a defect of the computation.

Theorem 4.1 (Analytic mechanism of the finite-size logarithmic slope bias near $\alpha_c = 1/2$). *Let $0 < \alpha < \frac{1}{2}$ and let $\tilde{K}_N(\alpha) = A[\frac{N^{1-2\alpha}}{1-2\alpha} + \zeta(2\alpha)]$ be the two-term Euler–Maclaurin form of $K_N(\alpha)$ from (3.2). Define the effective (local logarithmic) exponent $\gamma_{\text{eff}}(N; \alpha) = d \ln \tilde{K}_N / d \ln N$. Then*

$$\gamma_{\text{eff}}(N; \alpha) = \frac{1 - 2\alpha}{1 + (1 - 2\alpha) \zeta(2\alpha) N^{2\alpha-1}}, \quad (4.1)$$

and:

- $\gamma_{\text{eff}}(N; \alpha) > 1 - 2\alpha$ for every finite N (upward bias), and $\gamma_{\text{eff}}(N; \alpha) \downarrow 1 - 2\alpha$ as $N \rightarrow \infty$ at the rate $N^{2\alpha-1}$;
- the bias diverges relative to $1 - 2\alpha$ as $\alpha \uparrow \frac{1}{2}$: with $\epsilon = 1 - 2\alpha \rightarrow 0^+$, $(1 - 2\alpha) \zeta(2\alpha) N^{2\alpha-1} = -1 + \epsilon(\ln N + \gamma_{\text{EM}}) + O(\epsilon^2)$, and hence

$$\lim_{\alpha \uparrow 1/2} \gamma_{\text{eff}}(N; \alpha) = \frac{1}{\ln N + \gamma_{\text{EM}}}, \quad (4.2)$$

which is the exact local exponent of the critical law $A(\ln N + \gamma_{\text{EM}})$: the effective exponent is continuous across α_c , whereas $\gamma_{\text{th}} = 1 - 2\alpha$ jumps from 0^+ to the logarithmic regime;

- the true DKI satisfies $d \ln K_N / d \ln N = \gamma_{\text{eff}}(N; \alpha)(1 + O(N^{-1}) + O(c^2 N^{-2\alpha}))$ where $d/d \ln N$ acts on the Euler–Maclaurin expansion, so (4.1) is the leading finite-size law for the regression exponent.

Corollary 4.2 (Constant cancellation via the doubling tail estimator). *Under the standing assumption of §3 with $0 < \alpha < \frac{1}{2}$,*

$$D_{\text{tail}}(N) = A \frac{2^{1-2\alpha} - 1}{1 - 2\alpha} N^{1-2\alpha} \left(1 + O(N^{-1}) + O(c^2 N^{-2\alpha}) \right), \quad (4.3)$$

the constants $A\zeta(2\alpha)$ and $C_\infty(\alpha)$ cancelling identically. Consequently

$$\gamma_{\text{tail}}(N; \alpha) := \frac{d \ln D_{\text{tail}}(N)}{d \ln N} = 1 - 2\alpha + O(N^{-1}) + O(c^2 N^{-2\alpha}), \quad (4.4)$$

and the log-log regression of D_{tail} recovers the true exponent without the pole-induced bias of Theorem 4.1, uniformly in α up to the critical point.

The proofs rest on one elementary fact about ζ .

Lemma 4.3 (Sign and pole of ζ on $(0, 1)$). *For $0 < s < 1$, $\zeta(s) < 0$. As $s \uparrow 1$, $\zeta(s) = \frac{1}{s-1} + \gamma_{\text{EM}} + O(s-1)$; in particular $(1-s)\zeta(s) \rightarrow -1$.*

Proof. For $s > 0$ the Dirichlet eta function $\eta(s) = \sum_{n \geq 1} (-1)^{n-1} n^{-s}$ converges (alternating series with terms decreasing to 0) and is positive, being a sum of the positive pairs $(2m-1)^{-s} - (2m)^{-s}$. For $s > 1$, $\eta(s) = (1 - 2^{1-s})\zeta(s)$ by splitting ζ into even and odd n , and the identity persists for $s > 0$, $s \neq 1$, by analytic continuation. For $0 < s < 1$ the factor $1 - 2^{1-s}$ is negative, hence $\zeta(s) = \eta(s)/(1 - 2^{1-s}) < 0$. The Laurent expansion at $s = 1$ is classical [13, §2.1]: $\zeta(s) - \frac{1}{s-1}$ is entire and equals γ_{EM} at $s = 1$ (this is also (3.8) read against (3.7) as $s \rightarrow 1$). $\square$

Proof of Theorem 4.1. Differentiating $\ln \widetilde{K}_N$ with respect to $\ln N$, using $dN^{1-2\alpha}/d \ln N = (1 - 2\alpha)N^{1-2\alpha}$,

$$\gamma_{\text{eff}} = \frac{A N^{1-2\alpha}}{A \left[\frac{N^{1-2\alpha}}{1-2\alpha} + \zeta(2\alpha) \right]} = \frac{(1-2\alpha)N^{1-2\alpha}}{N^{1-2\alpha} + (1-2\alpha)\zeta(2\alpha)} = \frac{1-2\alpha}{1 + (1-2\alpha)\zeta(2\alpha)N^{2\alpha-1}},$$

which is (4.1). (i): by Lemma 4.3, $\zeta(2\alpha) < 0$ and $1 - 2\alpha > 0$, so the denominator is < 1 ; it is positive as soon as $N^{1-2\alpha} > (1-2\alpha)|\zeta(2\alpha)|$, which holds throughout the window considered (at $\alpha = 0.45$, $N = 10^3$: $N^{0.1} = 1.995 > 0.943$); $N^{2\alpha-1} \rightarrow 0$ gives the limit and the rate. (ii): put $s = 2\alpha = 1 - \epsilon$. By Lemma 4.3, $\epsilon \zeta(1 - \epsilon) = -1 + \gamma_{\text{EM}}\epsilon + O(\epsilon^2)$, and $N^{-\epsilon} = 1 - \epsilon \ln N + O(\epsilon^2)$; multiplying, $\epsilon \zeta(1 - \epsilon)N^{-\epsilon} = -1 + \epsilon(\ln N + \gamma_{\text{EM}}) + O(\epsilon^2)$, so the denominator of (4.1) is $\epsilon(\ln N + \gamma_{\text{EM}}) + O(\epsilon^2)$ and $\gamma_{\text{eff}} = \epsilon/(\epsilon(\ln N + \gamma_{\text{EM}}) + O(\epsilon^2)) \rightarrow 1/(\ln N + \gamma_{\text{EM}})$. That this is the local exponent of $A(\ln N + \gamma_{\text{EM}})$ is the computation $d \ln(\ln N + \gamma_{\text{EM}})/d \ln N = 1/(\ln N + \gamma_{\text{EM}})$. (iii): the terms of (3.2) omitted from $\widetilde{K}_N$ are $\frac{A}{2}N^{-2\alpha}$, of relative size N^{-1} against the leading term, and $R_N(\alpha) - C_\infty(\alpha)$ resp. $R_N(\alpha)$, of relative size $O(c^2 N^{-2\alpha})$ by Lemma 3.4 ($R_N \leq \varepsilon_c A H_N(4\alpha)$ against $AN^{1-2\alpha}/(1-2\alpha)$); the constant $C_\infty(\alpha)$, when present, adds to $A\zeta(2\alpha)$ and shifts the constant in (4.1) by the relative amount $C_\infty/(A\zeta(2\alpha)) = O(c^2)$, which is within the same error class. Differentiating the expansion term by term (it is an asymptotic expansion in powers of N , so differentiation with respect to $\ln N$ is legitimate) gives the claim. $\square$

Remark 4.4 (The $\alpha = 0.45$ anomaly, resolved). At $\alpha = 0.45$, $2\alpha = 0.9$ and $\zeta(0.9) = -9.4301140 \dots$ (from $\zeta(s) = \frac{1}{s-1} + \gamma_{\text{EM}} - \gamma_1(s-1) + \dots$ with $\gamma_1 = -0.0728158$: $-10 + 0.5772157 - 0.0072816 = -9.4300659$, the remaining Stieltjes terms contributing -4.8×10^{-5}). At the logarithmic centre $N_* = 10^4$ of the regression window $[10^3, 10^5]$, $N_*^{2\alpha-1} = 10^{-0.4} = 0.3981072$, so the denominator of (4.1) is $1 + 0.1 \times (-9.4301140) \times 0.3981072 = 1 - 0.3754185 = 0.6245815$ and

$$\gamma_{\text{eff}}(10^4; 0.45) = \frac{0.1000}{0.6245815} = 0.16011.$$

The regression value is $\gamma_{\text{num}}(0.45) = 0.16103$. The residual 0.0009 is the difference between the local exponent at the window centre and the least-squares slope over the whole window: fitting $\ln \widetilde{K}_N$ on the same sixty-point grid gives $\gamma_{\text{EM}}^{\text{fit}}(0.45) = 0.1613$ (Table 2), within 3×10^{-4} of γ_{num} . The same formula predicts the full column: 0.9001, 0.8003, 0.7007, 0.6016, 0.5037, 0.4080, 0.3167, 0.2327, 0.1601 against the measured 0.8995, 0.7998, 0.7004, 0.6016, 0.5039, 0.4085, 0.3174, 0.2336, 0.1610. The anomaly is therefore entirely accounted for by the negative, pole-dominated constant $A\zeta(2\alpha)$ competing with the slowly growing $AN^{1-2\alpha}/(1-2\alpha)$ inside a finite window. Note also the sanity check of (4.2): at $N = 10^4$ the critical local exponent

is $1/(9.2103 + 0.5772) = 0.1022$, so the sequence $\gamma_{\text{eff}}(10^4; \alpha)$ decreases continuously through 0.1601 at $\alpha = 0.45$ to 0.1022 at $\alpha = \frac{1}{2}$, while $1 - 2\alpha$ is already 0.1 at $\alpha = 0.45$ and 0 at α_c .

Proof of Corollary 4.2. Apply (3.2) at $2N$ and N and subtract. The N -independent terms $A\zeta(2\alpha)$ and, for $\alpha > \frac{1}{4}$, $C_\infty(\alpha)$ cancel exactly; the power terms give $\frac{A}{1-2\alpha}((2N)^{1-2\alpha} - N^{1-2\alpha}) = A\frac{2^{1-2\alpha}-1}{1-2\alpha}N^{1-2\alpha}$; the Euler–Maclaurin term $\frac{A}{2}((2N)^{-2\alpha} - N^{-2\alpha})$ is $O(N^{-2\alpha})$, i.e. relative $O(N^{-1})$; and the difference of the R -terms is, by Lemma 3.4, at most $\varepsilon_c A(H_{2N}(4\alpha) - H_N(4\alpha)) \leq \varepsilon_c AN \cdot N^{-4\alpha} = O(c^2 \cdot AN^{1-2\alpha} \cdot N^{-2\alpha})$, i.e. relative $O(c^2 N^{-2\alpha})$. Taking logarithms and differentiating the asymptotic expansion gives (4.4). $\square$

Remark 4.5. The last column of Table 2 confirms the corollary: γ_{tail} agrees with $1 - 2\alpha$ to within 5×10^{-4} at every α , including $\gamma_{\text{tail}}(0.45) = 0.1001$ where the direct estimator gives 0.1610. The small residuals at small α (0.8995 against 0.9) are the $O(c^2 N^{-2\alpha})$ term, which is largest when α is smallest ($c^2 N^{-0.1} \approx 4 \times 10^{-3}$ at $N = 10^4$). At the critical point itself the same estimator gives the universal constant of Proposition 3.2.

Remark 4.6 (The critical slope). For $\alpha = \frac{1}{2}$ the experiment regresses K_N on $\ln N$ over the same window and obtains the slope 0.0099992, against $A = 0.01$ predicted by (3.3). The deficit 8×10^{-7} is the derivative $-A/(2N)$ of the Euler–Maclaurin term $A/(2N)$, which is -5×10^{-6} at the left end of the window and is averaged down by the log-uniform weights; it is of the size predicted and of the predicted sign.

5. NUMERICAL VERIFICATION TABLES AND PHASE DIAGRAM ANALYSIS

Protocol. The script `exp01_marginal_dki_criticality.py` evaluates h_i from (2.2) in IEEE double precision for $i \leq 2 \times 10^5$, forms the cumulative sums K_N , the cumulative $\ln \Pi_N = \sum_i \ln(1 - h_i/2)$ (via `log1p`, so no underflow), and $D_{\text{tail}}(N) = K_{2N} - K_N$, for $\alpha \in \{0.25, 0.5, 0.75, 1\}$; it then regresses $\ln K_N$ on $\ln N$ (and K_N on $\ln N$ at $\alpha = \frac{1}{2}$) on a sixty-point log-uniform grid in $[10^3, 10^5]$ for $\alpha \in \{0.05, 0.10, \dots, 0.45\}$. The companion script `paper01_compute_tables.py` evaluates the analytic side: K_N^{EM} is (3.9) with H_N replaced by the four-term Euler–Maclaurin expansions (3.7)–(3.8) and $k \leq 6$; $\gamma_{\text{EM}}(10^4)$ is (4.1) at $N_* = 10^4$; $\gamma_{\text{EM}}^{\text{fit}}$ is the least-squares slope of $\ln \widetilde{K}_N$ on the regression grid; γ_{tail} is the least-squares slope of $\ln D_{\text{tail}}(N)$ on the same grid. All numbers below are copied verbatim from `exp01_results_benchmark.csv`, `exp01_results_scaling.csv` and their derived files `paper01_table1.csv`, `paper01_table2.csv`.

Acceptance benchmarks. The four target values fixed in advance for $N = 10^5$ were $K_{10^5} \approx 6.31, 0.121, 0.0262, 0.0166$ for $\alpha = 0.25, 0.5, 0.75, 1$; the measured values 6.3115, 0.12111, 0.026214, 0.016587 meet them with relative deviations 0.02%, 0.09%, 0.05%, 0.08%, i.e. to the three significant figures in which the targets were stated. The exponent target $\gamma_{\text{num}}(0.25) \approx 0.502$ is met by 0.5039 (Theorem 4.1 predicts 0.5037 at the window centre and 0.5042 on the window), and the critical-slope target 0.01 by 0.0099992.

Reading the tables. Three features of Table 1 deserve comment. First, the agreement between K_N and K_N^{EM} improves with N as a power, from 10^{-4} at $N = 10$ to 10^{-10} – 10^{-12} at $N = 10^5$, in accordance with the asymptotic character of the Euler–Maclaurin expansion; the data are exact to double precision, and the residual at large N is rounding. Second, the column $D_{\text{tail}}(N)$ separates the three phases without any fit: it grows as $N^{1/2}$ at $\alpha = 0.25$ (ratio ≈ 3.16 per decade), converges to 0.00693 at α_c , and decays as $N^{-1/2}$ resp. N^{-1} at $\alpha = 0.75$ resp. 1 (ratios ≈ 0.316 and 0.1 per decade). Third, Π_N at $N = 10^5$ equals 0.9412 at the critical point and 0.9870 in the nearest subcritical case: no finite- N observable distinguishes “will collapse to zero” from “has converged to 0.987” by its value alone—only the scaling with N does, which is why D_{tail} and not Π_N is the diagnostic proposed here. Table 2 shows that the pole-corrected exponent $\gamma_{\text{EM}}^{\text{fit}}$ predicts γ_{num} to within 3×10^{-4} over the whole range, and that γ_{tail} is unbiased to within 5×10^{-4} .

TABLE 1. Multi-scale benchmark data. The marginal drug Kakutani index K_N , the Kakutani affinity Π_N and the doubling tail increment $D_{\text{tail}}(N) = K_{2N} - K_N$ for $p = \frac{1}{2}$, $c = \frac{1}{10}$, together with the Euler–Maclaurin prediction K_N^{EM} of Lemma 3.4 (four-term expansions, $k \leq 6$) and the relative deviation $(K_N - K_N^{\text{EM}})/K_N$. The deviation at $N = 10$ is the truncation error of the asymptotic expansion, not of the data.

α	N	K_N	Π_N	$D_{\text{tail}}(N)$	K_N^{EM}	rel. dev.
0.25	10^1	0.05058144	0.974990	0.0258266	0.05060779	-5.2×10^{-4}
0.25	10^2	0.1865509	0.910883	0.082783	0.1865517	-4.5×10^{-6}
0.25	10^3	0.6189526	0.733762	0.262012	0.6189526	-4.3×10^{-8}
0.25	10^4	1.986677	0.370293	0.828499	1.986677	-4.2×10^{-10}
0.25	10^5	6.311486	0.042600	2.619799	6.311486	-3.8×10^{-12}
0.50	10^1	0.02948662	0.985345	0.00669352	0.02948666	-1.1×10^{-6}
0.50	10^2	0.05208138	0.974275	0.00690715	0.05208138	-7.5×10^{-10}
0.50	10^3	0.07506343	0.963144	0.00692903	0.07506343	2.5×10^{-11}
0.50	10^4	0.09808489	0.952121	0.00693123	0.09808489	1.9×10^{-11}
0.50	10^5	0.1211103	0.941222	0.00693145	0.1211103	1.6×10^{-11}
0.75	10^1	0.0201059	0.989982	0.00175387	0.0201138	-3.9×10^{-4}
0.75	10^2	0.02428183	0.987917	0.000582569	0.02428186	-1.0×10^{-6}
0.75	10^3	0.02564455	0.987244	0.00018514	0.02564455	-3.0×10^{-9}
0.75	10^4	0.02607685	0.987031	5.85754×10^{-5}	0.02607685	6.2×10^{-11}
0.75	10^5	0.0262136	0.986964	1.85241×10^{-5}	0.0262136	7.2×10^{-11}
1.00	10^1	0.01563567	0.992199	0.000463986	0.015639	-2.1×10^{-4}
1.00	10^2	0.01648787	0.991776	4.96265×10^{-5}	0.01648787	-2.0×10^{-7}
1.00	10^3	0.01657738	0.991732	4.99625×10^{-6}	0.01657738	-8.8×10^{-11}
1.00	10^4	0.01658637	0.991727	4.99963×10^{-7}	0.01658637	1.1×10^{-10}
1.00	10^5	0.01658727	0.991727	4.99996×10^{-8}	0.01658727	1.1×10^{-10}

Source files: `exp01_results_benchmark.csv` (columns 3–5), `paper01_table1.csv` (columns 6–7).

TABLE 2. Scaling exponents. Asymptotic exponent $\gamma_{\text{th}} = 1 - 2\alpha$; the pole-dominated constant $\zeta(2\alpha)$; the least-squares exponent γ_{num} of $\ln K_N$ on $\ln N$ over $[10^3, 10^5]$; the Euler–Maclaurin local exponent $\gamma_{\text{EM}}(10^4)$ of (4.1) at the window centre; the least-squares exponent $\gamma_{\text{EM}}^{\text{fit}}$ of the analytic $\ln \widetilde{K}_N$ on the same grid; and the doubling-tail exponent γ_{tail} of Corollary 4.2. Source files: `exp01_results_scaling.csv`, `paper01_table2.csv`.

α	$\gamma_{\text{th}} = 1 - 2\alpha$	$\zeta(2\alpha)$	γ_{num}	$\gamma_{\text{EM}}(10^4)$	$\gamma_{\text{EM}}^{\text{fit}}$	γ_{tail}
0.05	0.9000	−0.6030	0.8995	0.9001	0.9001	0.8995
0.10	0.8000	−0.7339	0.7998	0.8003	0.8003	0.7996
0.15	0.7000	−0.9046	0.7004	0.7007	0.7009	0.6998
0.20	0.6000	−1.1348	0.6016	0.6016	0.6019	0.5999
0.25	0.5000	−1.4604	0.5039	0.5037	0.5042	0.5000
0.30	0.4000	−1.9527	0.4085	0.4080	0.4087	0.4000
0.35	0.3000	−2.7784	0.3174	0.3167	0.3176	0.3000
0.40	0.2000	−4.4375	0.2336	0.2327	0.2339	0.2000
0.45	0.1000	−9.4301	0.1610	0.1601	0.1613	0.1001

Critical slope of K_N on $\ln N$ at $\alpha = 0.50$: measured 0.0099992, predicted $A = 0.01$.

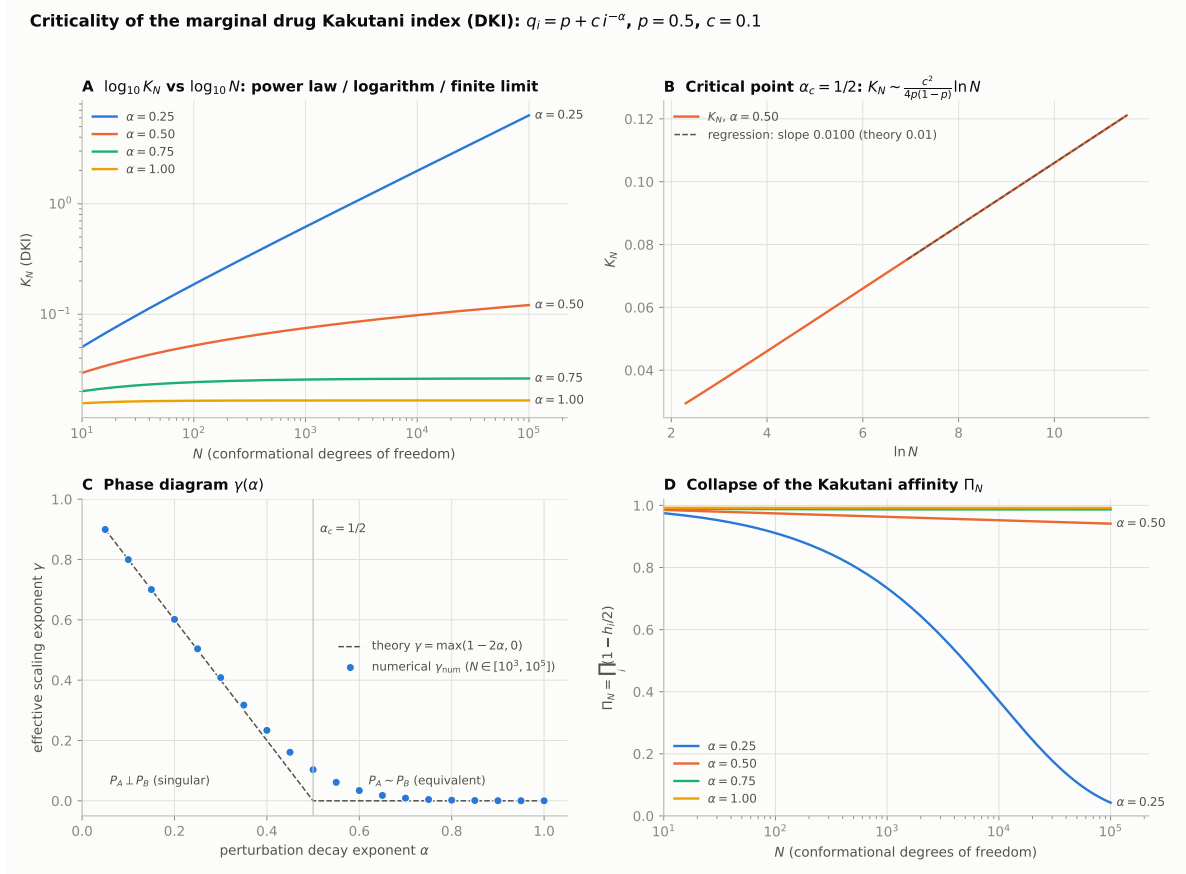


FIGURE 1. Criticality of the marginal drug Kakutani index for $q_i = \frac{1}{2} + \frac{1}{10}i^{-\alpha}$ (figures/fig01_marginal_dki_phase_transition.pdf, with a PNG copy alongside). **A.** K_N against N on doubly logarithmic axes for $\alpha = 0.25, 0.5, 0.75, 1$. The supercritical curve ($\alpha = 0.25$) is a straight line of slope $\rightarrow \frac{1}{2}$, the signature of $K_N \sim 2A\sqrt{N}$; the critical curve bends downward continuously, as a logarithm must on these axes; the subcritical curves saturate at the finite limits $K_\infty(0.75) = 0.026277$ and $K_\infty(1) = 0.016587$ of (3.5). The vertical order of the curves at fixed N is the order of $H_N(2\alpha)$. **B.** K_N against $\ln N$ at $\alpha = \alpha_c$: a straight line of slope $A = 0.01$ (regression 0.0099992), the intercept being $A\gamma_{\text{EM}} + C_\infty(\frac{1}{2}) = 0.005981$; the strict linearity over four decades is the content of (3.3), and its slope is the Fisher-information prefactor $c^2/(4p(1-p))$. **C.** The phase diagram: the measured regression exponent $\gamma_{\text{num}}(\alpha)$ (points) against the asymptotic $\max(1 - 2\alpha, 0)$ (dashed). The points lie systematically above the line for $\alpha < \frac{1}{2}$ and approach it from above as $\alpha \rightarrow 0$; the excess is $\gamma_{\text{eff}} - (1 - 2\alpha)$ of Theorem 4.1, growing with $|\zeta(2\alpha)|$ as the pole at $2\alpha = 1$ is approached, and it does not vanish at α_c but joins continuously to $1/(\ln N + \gamma_{\text{EM}})$ by (4.2). For $\alpha > \frac{1}{2}$ the points decay to 0 as $N^{1-2\alpha}$, the rate of saturation in (3.5). The vertical line marks the Kakutani boundary: to its left the infinite ensembles are mutually singular, to its right equivalent. **D.** The Kakutani affinity $\Pi_N = \prod_{i \leq N} (1 - h_i/2)$. At $\alpha = 0.25$ it collapses to 0.0426 at $N = 10^5$ along $\exp(-K_N/2)$; at α_c it decays as the power $N^{-A/2} = N^{-0.005}$ (from 0.9853 at $N = 10$ to 0.9412 at $N = 10^5$) and will reach zero only logarithmically slowly; at $\alpha = 0.75$ and 1 it has converged, to within 3×10^{-5} and 10^{-7} respectively, to its positive limits $\Pi_\infty = 0.98693$ and 0.99173 . Every finite- N affinity is strictly positive: the singularity is a statement about the limit, as Theorem 2.6 requires.

6. ASSESSMENT AND BRIDGE TO PAPER II

We separate what is established from what is not.

Established unconditionally. Within the product model of Definition 2.1: the product formula and the logarithmic sandwich for the affinity (Lemma 2.5); the exact quartic expansion (2.4) and the even closed form (2.5) at $p = \frac{1}{2}$ (Lemma 2.3); the trichotomy with its constants (Theorem 3.1), the critical doubling identity (Proposition 3.2), the effective-exponent law (4.1) with its continuity (4.2) through the critical point (Theorem 4.1), and the unbiasedness of the doubling estimator (Corollary 4.2). The numerical benchmarks of Section 5 are consistent with these statements to the precision of double arithmetic; they verify the theorems, they are not needed for them.

Conditional or incomplete. The identification of the physical ensemble with a *product* measure is a modelling assumption, and it is the one that Paper II removes. The power law $q_i = p + ci^{-\alpha}$ is a model of allosteric decay along a one-dimensional index; a residue graph with its own metric would replace $i^{-\alpha}$ by a decay in graph distance and $H_N(2\alpha)$ by a growth function of the graph, and the trichotomy would be inherited from the growth rate of that function. The critical value $\alpha_c = \frac{1}{2}$ is therefore specific to $\mathbb{Z}$ -like indexing; what is model-independent is the mechanism (square-summability of the local Hellinger coordinates) and the diagnostic (the doubling tail). Finally, $N \rightarrow \infty$ is an idealisation: a protein has finitely many residues, and Lemma 2.5 shows that every Π_N is strictly positive. The theorems say how the finite- N observables scale, and Section 4 shows that the scaling, not the value, is what can be read off.

Relation to the literature. Kakutani’s theorem [8] and its Gaussian form [5, 6] are classical; the Hellinger affinity and its relation to Fisher information are in [7, 2, 9]. The transport from [4] is at the level of the criterion (the dictionary of §1); nothing here uses the operator-algebraic structure of the Bost–Connes system, and conversely nothing there uses the Bernoulli structure. The ensemble view of allostery [11, 12] supplies the physical premise that function is a property of the joint distribution; we have not seen the Kakutani dichotomy used in that literature, and a literature check remains advisable.

TABLE 3. Assessment of Paper I.

Item	Status	Where
DKI, affinity, doubling tail; product formula $\Pi_N = \prod(1 - h_i/2)$	established	Def. 2.2, Lemma 2.5
Quartic expansion of h ; vanishing of odd terms and closed form at $p = \frac{1}{2}$	established	Lemma 2.3
Phase trichotomy, $\alpha_c = \frac{1}{2}$, constants $A\zeta(2\alpha)$, $C_\infty(\alpha)$	established (product model)	Thm. 3.1
Critical doubling constant $A \ln 2$	established	Prop. 3.2
Effective exponent $\gamma_{\text{eff}}(N; \alpha)$; near-critical bias = pole of ζ	established	Thm. 4.1, Rem. 4.4
Unbiased doubling estimator	established	Cor. 4.2
Benchmarks at $N = 10^5$; agreement with Euler– Maclaurin to 10^{-10}	verified numerically	Tables 1, 2
Independence of local switches	modelling assump- tion	§6, Paper II
Identification of α with a structural decay length	open	Problem (2)
Covariance-driven transition with all marginals fixed	open	Problem (1), Paper II

Open problems and the bridge to Paper II.

- (1) *Covariance dominance.* Residue fluctuations in a real protein are not independent: hydrogen-bond networks, disulfide bridges and the hydrophobic core couple them through a covariance matrix C_{ij} . Consider the Gaussian ensembles $\mathcal{N}(\mu_A, C_A)$ and $\mathcal{N}(\mu_B, C_B)$ on $\mathbb{R}^N$, for which the Bhattacharyya distance $D_B = D_{\text{mean}} + D_{\text{cov}}$ splits into a mean part, which is the continuous analogue of the DKI studied here, and a covariance part

$$D_{\text{cov}} = \frac{1}{2} \ln \frac{\det \frac{1}{2}(C_A + C_B)}{\sqrt{\det C_A \det C_B}}.$$

Suppose the mean shifts $\Delta\mu_i$ lie in the subcritical regime of Theorem 3.1—or vanish identically, $\Delta\mu_i \equiv 0$ and $\Delta\text{RMSF}_i \equiv 0$, so that the marginal index is $K_N \equiv 0$. Can the off-diagonal re-arrangement of C_{ij} alone drive the Feldman–Hájek dichotomy into the singular regime, i.e. $D_{\text{cov}} \rightarrow \infty$ as $N \rightarrow \infty$? The finite-dimensional Gaussian measures are always equivalent, so the question is one of scaling of D_{cov} with N , and the doubling-tail diagnostic of Corollary 4.2 applies verbatim. Paper II defines the correlated Kakutani index (CKI) and examines the regime—suggested by preliminary computation and to be established there—in which the covariance term carries the overwhelming share of D_B while every marginal is unchanged.

- (2) *What is α ?* Relate the decay exponent to a measurable property of the residue-interaction graph (its growth function or spectral dimension), so that $\alpha \leq \alpha_c$ becomes a structural prediction rather than a fitted parameter.
- (3) *Beyond $p = \frac{1}{2}$.* For $p \neq \frac{1}{2}$ the cubic term of (2.4) does not vanish and the sign of δ_i matters; the trichotomy persists (the quadratic term still dominates) but the constants acquire odd contributions $\propto c^3 H_N(3\alpha)$. Determine whether the maximal-entropy point $p = \frac{1}{2}$ is selected physically by the ground state or is merely convenient.
- (4) *The arithmetic side.* The dictionary of §1 suggests that the finite-size theory of Section 4—the pole of ζ as the source of apparent exponents—has a counterpart for the obstruction group $\Xi_\beta(K, S)$ of [4] when S is a finite truncation of an infinite set; whether the “doubling” estimator has an arithmetic meaning is not known to us.

Reproducibility. All results are generated by `code/exp01_marginal_dki_criticality.py` (data, figure, acceptance tests) and `code/paper01_compute_tables.py` (analytic constants, Tables 1–2); both run in seconds with NumPy, Matplotlib and mpmath. The paper and its L^AT_EX source (`paper/`), the raw data (`data/`), the figure (`figures/`) and the derived tables (`results/`) are archived under DOI [10.5281/zenodo.23005921](https://doi.org/10.5281/zenodo.23005921) and maintained in the repository `kakutani-statistical-pharmacology-I`.

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